OOPJ ASSIGNMENT 13

Name – Venya Kejriwal

Roll Number – 24155517

1. Life Cycle of Thread

```
class LifeCycleThread extends Thread {
    public void run() {
        System.out.println("Thread is running...");
    }}
class ThreadLifeCycleDemo {
    public static void main(String[] args) throws InterruptedException {
        LifeCycleThread t = new LifeCycleThread();
        System.out.println("State after creation: " + t.getState());
        t.start();
        System.out.println("State after start(): " + t.getState());
        t.join();
        System.out.println("State after completion: " + t.getState());
        System.out.println("VENYA KEJRIWAL - 24155517");
    }}
```

OUTPUT

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\l
State after creation: NEW
State after start(): RUNNABLE
Thread is running...
State after completion: TERMINATED
VENYA KEJRIWAL - 24155517
```

2. Synchronized Method Using Anonymous Class

```
class syn {
    int count = 0;
    synchronized void increment() {
        count++;
        System.out.println("Count: " + count);
    }}
class SynchronizedThreadDemo {
    public static void main(String[] args) {
        syn c = new syn();
        new Thread() {
            public void run() {
                for (int i = 0; i < 5; i++) {
                    c.increment();
                }}}.start();
        System.out.println("VENYA KEJRIWAL - 24155517");
    }}
```

OUTPUT

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\l
VENYA KEJRIWAL - 24155517
Count: 1
Count: 2
Count: 3
Count: 4
Count: 5
```

3. Static Synchronized Block

```java
class StaticCounter {
    static int count = 0;
    static synchronized void increment() {
        count++;
        System.out.println("Static Count: " + count);
    }}
class StaticSyncDemo {
    public static void main(String[] args) {
        new Thread() {
            public void run() {
                for (int i = 0; i < 5; i++) {
                    StaticCounter.increment();
                }}}.start();
        System.out.println("VENYA KEJRIWAL - 24155517");
    }}
```

OUTPUT

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\l
VENYA KEJRIWAL - 24155517
Static Count: 1
Static Count: 2
Static Count: 3
Static Count: 4
Static Count: 5
```

4. Deadlock

```java
class DeadlockExample {
    public static void main(String[] args) {
        final Object lock1 = new Object();
        final Object lock2 = new Object();
        new Thread() {
            public void run() {
                synchronized(lock1) {
                    try { Thread.sleep(50); } catch(Exception e) {}
                    synchronized(lock2) {
                        System.out.println("Thread 1 finished");
                    }}}}.start();
        new Thread() {
            public void run() {
                synchronized(lock2) {
                    try { Thread.sleep(50); } catch(Exception e) {}
                    synchronized(lock1) {
                        System.out.println("Thread 2 finished");
                    }}}}.start();
        System.out.println("VENYA KEJRIWAL - 24155517");
    }}
```

OUTPUT

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\
VENYA KEJRIWAL - 24155517
```

5. Deadlock Avoiding

```java
class NoDeadlockExample {
    public static void main(String[] args) {
        final Object lock1 = new Object();
        final Object lock2 = new Object();
```

```java
        new Thread() {
            public void run() {
                synchronized(lock1) {
                    try { Thread.sleep(50); } catch(Exception e) {}
                    synchronized(lock2) {
                        System.out.println("Thread 1 finished safely");
                    }}}}.start();
        new Thread() {
            public void run() {
                synchronized(lock1) {  // same order
                    try { Thread.sleep(50); } catch(Exception e) {}
                    synchronized(lock2) {
                        System.out.println("Thread 2 finished safely");
                    }}}}.start();
        System.out.println("VENYA KEJRIWAL - 24155517");
    }}
```

OUTPUT

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\l
VENYA KEJRIWAL - 24155517
Thread 1 finished safely
Thread 2 finished safely
```

6. Interrupted Thread Code

```java
class inter extends Thread {
    public void run() {
        for(int i=1;i<=5;i++){
            System.out.println("Count: " + i);
            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {
                System.out.println("Thread interrupted but continues...");
            }}
        System.out.println("Thread finished normally");
    }}
public class InterruptDemo {
    public static void main(String[] args){
        inter t = new inter();
        t.start();
        try { Thread.sleep(2000); } catch(Exception e) {}
        t.interrupt();
        System.out.println("VENYA KEJRIWAL - 24155517");
    }}
```

OUTPUT

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\lib\
Count: 1
Count: 2
VENYA KEJRIWAL - 24155517
Thread interrupted but continues...
Count: 3
Count: 4
Count: 5
Thread finished normally
```

7. Is Interrupt Thread

```java
class CheckInterrupt extends Thread {
    public void run() {
```

```java
        for(int i=1;i<=5;i++){
            System.out.println("Count: " + i);
            if(isInterrupted()) {
                System.out.println("Detected interrupt using isInterrupted()");
            }
            try { Thread.sleep(1000); }
            catch(InterruptedException e) {
                System.out.println("InterruptedException caught");
                System.out.println("Thread interrupted status using interrupted(): " +
Thread.interrupted());}}}}
class InterruptCheckDemo {
    public static void main(String[] args){
        CheckInterrupt t = new CheckInterrupt();
        t.start();
        try { Thread.sleep(2500); } catch(Exception e) {}
        t.interrupt();
    }}
```

OUTPUT

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\lib
Count: 1
Count: 2
Count: 3
VENYA KEJRIWAL - 24155517
InterruptedException caught
Thread interrupted status using interrupted(): false
Count: 4
Count: 5
```

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